GOV51 PSET4

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https://github.com/Steffny2004/gov51-ps4

1 Setup and Exploration (10 points)

1.1 Load packages and data

Load the `tidyverse`, `estimatr`, and `modelsummary` packages. Then load the dataset from `data/card1995.csv`.

```
library(tidyverse)
library(estimatr)
library(modelsummary)
card <- read_csv("data/raw/card1995.csv")
```

1.2 Describe the data

How many observations are in the dataset? What is the unit of observation?

```
nrow(card)
```

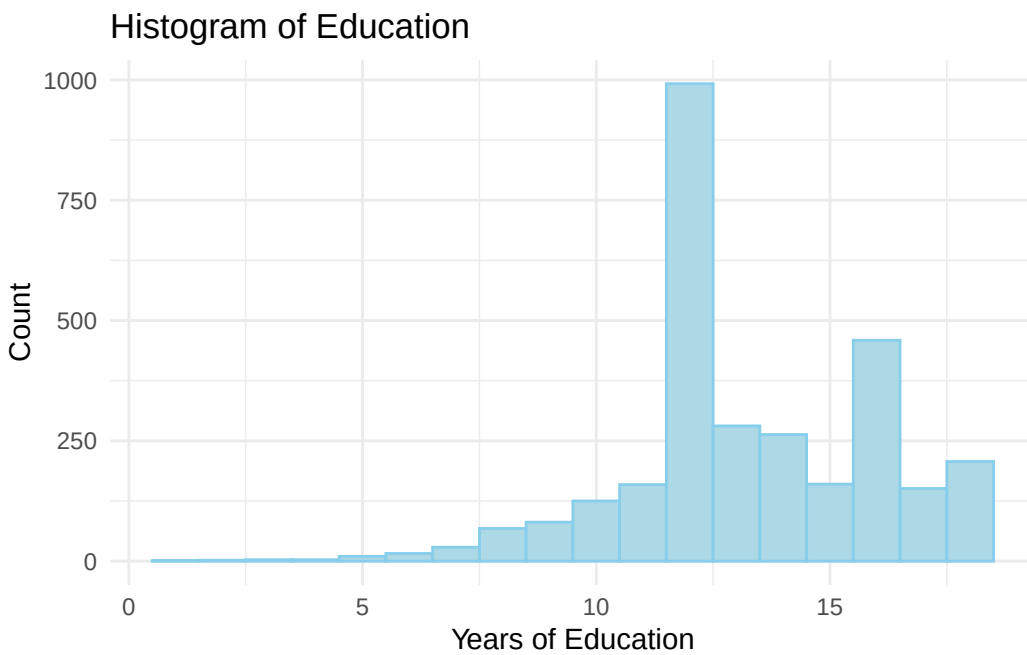
```
[1] 3010
```

There are 3010 observations in the dataset, with the unit of observation being individual men from the 1976 U.S.

1.3 Distribution of years of education

Create a histogram of `educ`. What is the mean? Are there any unusual spikes? What might explain them?

```r
# Plot histogram
ggplot(data = card, mapping = aes(x = educ)) +
  geom_histogram(binwidth = 1, color = "skyblue", fill = "lightblue") +
  labs(
    title = "Histogram of Education",
    x = "Years of Education",
    y = "Count") +
  theme_minimal()
```




```r
# Compute mean
mean(card$educ)
```

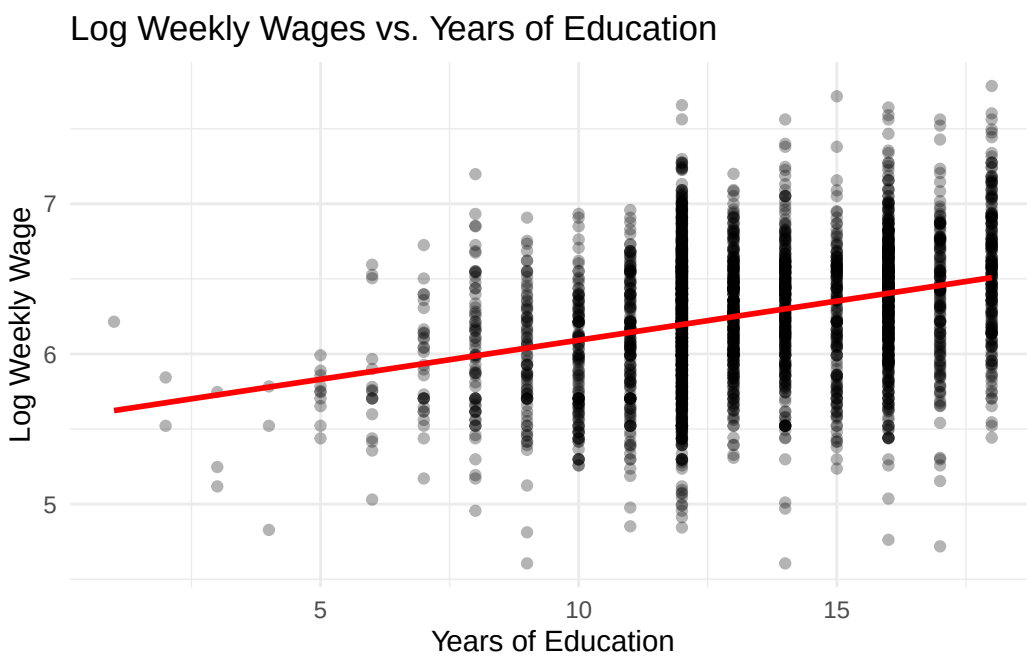
```
[1] 13.26346
```

The mean level of education in the sample is 13.26 years. Looking at the histogram, there are spikes at 12 and 16 years, which makes sense as they correspond to completing high school and college.

1.4 The naïve relationship

Create a scatter plot of `lwage` (y-axis) against `educ` (x-axis). Add a linear regression line. Based on this plot, does more education appear to be associated with higher wages?

```r
ggplot(card, aes(x = educ, y = lwage)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(title = "Log Weekly Wages vs. Years of Education",
       x = "Years of Education",
       y = "Log Weekly Wage") +
  theme_minimal()
```



The scatter plot seems to show a positive association between years of education and log weekly wages. The fitted regression line slopes upward, which suggests that individuals with more education tend to earn higher wages, although this is a purely correlative statement, not causal.

1.5 Why OLS might be misleading

The OLS regression of `lwage` on `educ` might not give us the causal effect of education. Name two specific reasons why people with more education might earn more *even if education itself had no effect.*

1. People with more education may earn more due to innate ability, meaning that they are more likely to either obtain additional schooling, or earn higher wages due to a higher baseline of ability, regardless of education.

2. People with more education may earn more due to having an advantageous family background, meaning that they come from wealthier or more educated families that allow them to either attend more schooling wihtout financial worry, or may have better networks and connection to job opportunities.

2 The Instrument: College Proximity (25 points)

2.1 What the instrument measures

The variable `nearc4` equals 1 if the respondent grew up near a 4-year college. What share of respondents in this sample grew up near a 4-year college?

```
sum(card$nearc4) / nrow(card)
```

```
[1] 0.6820598
```

68.2% of the respondents in this sample grew up near a 4-year college.

2.2 Relevance: does proximity affect education?

The first condition for a valid instrument is relevance: the instrument must be correlate with the endogenous variable (education).

Run the following regression — the **first stage** — and report the coefficient on `nearc4` along with its standard error and t-statistic.

```
first_stage <- lm_robust(educ ~ nearc4 + black + smsa + south + exper + expersq + married,
                         data = card, se_type = "HC2")
summary(first_stage)
```

```
Call:
lm_robust(formula = educ ~ nearc4 + black + smsa + south + exper +
    expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
            Estimate Std. Error  t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept) 16.94691   0.160943 105.2976 0.000e+00 16.631341 17.262481 2995
nearc4       0.32711   0.080554   4.0607 5.017e-05  0.169164  0.485059 2995
black       -0.94529   0.088599 -10.6694 4.142e-26 -1.119011 -0.771571 2995
smsa         0.42333   0.085211   4.9680 7.144e-07  0.256251  0.590405 2995
south       -0.30018   0.078571  -3.8204 1.359e-04 -0.454235 -0.146117 2995
exper       -0.43245   0.032388 -13.3522 1.520e-39 -0.495950 -0.368941 2995
expersq      0.00141   0.001707   0.8259 4.089e-01 -0.001937  0.004757 2995
```

```
married     -0.07441   0.017670  -4.2109 2.618e-05 -0.109055 -0.039761 2995

Multiple R-squared:  0.4775 ,   Adjusted R-squared:  0.4763
F-statistic: 523.3 on 7 and 2995 DF,  p-value: < 2.2e-16
```

Interpret the coefficient on `nearc4` in a sentence. Does the sign make sense?

The coefficent on `near4c` is 0.33 with an SE of 0.08 and a t-statistic of 4.06, indicating that this instrument is singificant at the $\alpha = 0.05$ level. The coefficient of 0.33 means that growing up near a 4-year college is associated with about 0.33 additional years of education, holding other factors constant. This makes sense that this results in a positive effect on education, since having proximity to college can be an indicator for other aspects of an individuals' environment and upbringing that encourages them to continue more schooling.

2.3 Checking instrument strength

Report the t-statistic on `nearc4` from the first-stage regression. Compute the first-stage F-statistic as $F = t^2$. Is your instrument strong? (Rule of thumb: F > 10 suggests a strong instrument.) Why does instrument strength matter for the reliability of the 2SLS estimate?

The t-statistic on `near4c` is 4.0607, so the F-statistics is $F = (4.0607)^2 = 16.489$. Since our F-statistic is greater than 10, we consider this to be a strong instrument. Instrument strength is important for the reilability of the 2SLS estimate because weak instruments can lead to biased and imprecise 2SLS estimates, making causal inference unreliable.

2.4 Exclusion restriction

The second condition for a valid instrument is the exclusion restriction: college proximity must affect wages only through its effect on education — not through any other channel.

(a) State the exclusion restriction formally: what must be true about the relationship between `nearc4` and `lwage` once we control for education?

The exclusion restriction requires that once we control for education (and the other included covariates), `nearc4` should have no direct effect on wages. So, living near a 4-year college should affect `lwage` only because it changes `educ`, not through any other pathway.

(b) Can you think of a way the exclusion restriction might be violated — that is, a reason why growing up near a college might affect your wages for reasons other than more education? Be specific.

The exclusion restriction might be violated if we believe that growing up near a college may mean growing up in a more urban or economically advantaged area, which would position an individual to have more educated family and peers, more networks, and more opportunities. If this is true, then college proximity could raise wages even apart from education.

2.5 Balance check

If the instrument is as-good-as-randomly assigned, respondents who grew up near a college should look similar on pre-treatment characteristics to those who did not. Compute the mean black, smsa, and south for the two groups (`nearc4 == 0` and `nearc4 == 1`). Are the groups balanced? What does any imbalance tell you about the instrument?

```
card %>%
  group_by(nearc4) %>%
  summarize(
    mean_black = mean(black),
    mean_smsa  = mean(smsa),
    mean_south = mean(south))
```

```
# A tibble: 2 x 4
  nearc4 mean_black mean_smsa mean_south
   <dbl>      <dbl>     <dbl>      <dbl>
1      0      0.280     0.479      0.563
2      1      0.212     0.822      0.329
```

From the table, we can see that the groups are not fully balanced. Individuals who grew up near a college (`nearc4 == 1`) are more likely to live in metropolitan areas and less likely to live in the South, which suggests that the instrument is correlated with geographic and socioeconomic factors rather than being randomly assigned. These imbalances could mean our exclusion restriction is violated, since these characteristics can directly affect wages.

3 The Wald Estimator (15 points)

The simplest IV estimate is the Wald estimator: the reduced-form effect divided by the first-stage effect.

$$\hat{\tau}_{Wald} = \frac{\text{Reduced Form}}{\text{First Stage}} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]}$$

3.1 Compute the reduced form

Run the reduced form regression: regress `lwage` directly on `nearc4` (without `educ`, but including the same controls as the first stage). Report the coefficient on `nearc4` and interpret it: what does it tell us about how proximity to a college relates to wages?

```
reduced_form <- lm_robust(lwage ~ nearc4 + black + smsa + south + exper + expersq + married,
                          data = card,se_type = "HC2")

summary(reduced_form)
```

```
Call:
lm_robust(formula = lwage ~ nearc4 + black + smsa + south + exper +
    expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
             Estimate Std. Error t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept)  6.104784  0.0369400 165.262 0.000e+00  6.032354  6.177214 2995
nearc4       0.040854  0.0160993   2.538 1.121e-02  0.009287  0.072421 2995
black       -0.236173  0.0183381 -12.879 5.575e-37 -0.272130 -0.200217 2995
smsa         0.196879  0.0166562  11.820 1.527e-31  0.164220  0.229537 2995
south       -0.146503  0.0163740  -8.947 6.243e-19 -0.178609 -0.114398 2995
exper        0.040916  0.0067800   6.035 1.787e-09  0.027622  0.054210 2995
expersq     -0.001789  0.0003228  -5.543 3.240e-08 -0.002422 -0.001156 2995
married     -0.038826  0.0038392 -10.113 1.155e-23 -0.046353 -0.031298 2995

Multiple R-squared:  0.2171 ,   Adjusted R-squared:  0.2152
F-statistic: 128.2 on 7 and 2995 DF,  p-value: < 2.2e-16
```

The reduced form regression shows that growing up near a 4-year college is associated with an increase of about 0.04 in log weekly wages with an SE of 0.03 and a t-statistic of 2.538, indicating its significance at the $\alpha = 0.05$ level. This suggests that individuals who lived near a college earn about 4% higher wages on average.

3.2 Calculate the Wald estimator by hand

Using the first-stage coefficient from Section 2.2 and the reduced-form coefficient from Section 3.1, compute the Wald estimator by hand. Show your calculation.

$$\hat{\tau}_{Wald} = \frac{[\text{reduced form coeff on nearc4}]}{[\text{first stage coeff on nearc4}]}$$

$$\hat{\tau}_{Wald} = \frac{0.040854}{0.32711}$$

$$\hat{\tau}_{Wald} = 0.1248938$$

3.3 Interpret the Wald estimator

Interpret your Wald estimate in a sentence. What does it say about the effect of one additional year of education on wages?

The Wald estimator says that an additional year of education raises log weekly wages by about 0.125, or about 12.5%. This suggests a large positive return to schooling for individuals whose education is affected by proximity to a college.

4 OLS vs. 2SLS (35 points)

4.1 OLS estimate

Run OLS of `lwage` on `educ` and the controls. Report the coefficient on `educ`.

```r
ols <- lm_robust(lwage ~ educ + black + smsa + south + exper + expersq + married,
                 data = card, se_type = "HC2")
summary(ols)
```

```
Call:
lm_robust(formula = lwage ~ educ + black + smsa + south + exper +
    expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
             Estimate Std. Error t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept)  4.902202   0.071286  68.768 0.000e+00  4.762427  5.041977 2995
educ         0.071498   0.003611  19.799 4.317e-82  0.064418  0.078579 2995
black       -0.168717   0.017400  -9.696 6.533e-22 -0.202835 -0.134600 2995
smsa         0.172431   0.015065  11.446 1.018e-29  0.142891  0.201970 2995
south       -0.127692   0.015174  -8.415 5.996e-17 -0.157444 -0.097939 2995
exper        0.071813   0.006738  10.658 4.655e-26  0.058602  0.085024 2995
expersq     -0.001889   0.000315  -5.998 2.243e-09 -0.002507 -0.001271 2995
married     -0.033550   0.003580  -9.373 1.348e-20 -0.040569 -0.026532 2995

Multiple R-squared:  0.3133 ,   Adjusted R-squared:  0.3117
F-statistic: 208.4 on 7 and 2995 DF,  p-value: < 2.2e-16
```

The coefficient on `educ` is 0.071498, meaning that each additional year of education is associated with about a 0.07 increase in log weekly wages, or roughly a 7% increase in wages.

4.2 2SLS estimate

Run the 2SLS regression using `iv_robust()` from `estimatr`. The formula syntax separates the structural equation (left of |) from the first-stage instruments (right of |). Report the coefficient on `educ`. Does it match your Wald estimate from Section 3.2?

```r
iv <- iv_robust(lwage ~ educ + black + smsa + south + exper + expersq + married |
                  nearc4 + black + smsa + south + exper + expersq + married,
                data = card, se_type = "HC2")

summary(iv)
```

```
Call:
iv_robust(formula = lwage ~ educ + black + smsa + south + exper +
    expersq + married | nearc4 + black + smsa + south + exper +
    expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
             Estimate Std. Error t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept)  3.988229  0.8399956   4.748 2.152e-06  2.341203  5.635256 2995
educ         0.124893  0.0490790   2.545 1.099e-02  0.028661  0.221125 2995
black       -0.118113  0.0493318  -2.394 1.672e-02 -0.214841 -0.021385 2995
smsa         0.144008  0.0305021   4.721 2.452e-06  0.084201  0.203815 2995
south       -0.109013  0.0230567  -4.728 2.371e-06 -0.154222 -0.063805 2995
exper        0.094925  0.0223208   4.253 2.176e-05  0.051160  0.138691 2995
expersq     -0.001965  0.0003466  -5.670 1.565e-08 -0.002645 -0.001286 2995
married     -0.029533  0.0051528  -5.731 1.096e-08 -0.039636 -0.019429 2995

Multiple R-squared:  0.2587 ,   Adjusted R-squared:  0.257
F-statistic:   135 on 7 and 2995 DF,  p-value: < 2.2e-16
```

The coefficent on `educ` is 0.124893, which does closely match the hand-calcualted Wald estimate of 0.1248938 from Section 3.2.

4.3 Side-by-side comparison

Display both the OLS and 2SLS estimates in a single regression table using `modelsummary()`. Keep only the rows for `educ` and the model statistics (N, R^2).

```r
modelsummary(list("OLS" = ols, "2SLS" = iv),
coef_map = c("educ" = "Years of Education"),
gof_map = c("nobs", "r.squared"))
```

	OLS	2SLS
Years of Education	0.071	0.125
	(0.004)	(0.049)
Num.Obs.	3003	3003
R2	0.313	0.259

4.4 Why do OLS and 2SLS differ?

The OLS and 2SLS estimates of the return to education are likely to differ. Answer the following:

(a) Which is larger — OLS or 2SLS?

The 2SLS estimate of the return to education (0.125) is larger than the OLS estimate (0.071). This difference suggests that OLS may be biased, potentially due to measurement error or other confounding factors. The IV estimate identifies the variation in education due tp college proximity, which leads to a larger estimated return overall.

(b) Card (1995) interprets the finding that 2SLS > OLS as evidence of downward bias in OLS. What could cause OLS to underestimate the return to education? *(Hint: think about measurement error in* `educ`*, or about who the compliers are.)*

OLD might underestimate the return to education because of measurement error in `educ`, meaning that if years of education are measured with error through misreporting or rounding, then the estimated coefficient for OLS might be biased towards 0 and make the return to education look smaller than it really is. Also, the 2SLS estimate identifies the effect of education for compliers, which are individuals whose education are affected by proximity to a college. As a result, these individuals may have higher returns to education, causing the IV estimate to exceed the OLS estimate.

(c) Which estimate is more credible as a causal estimate of the effect of education on wages? Explain briefly.

The 2SLS estimate is more credible as a causal estimate because it uses an instrumental variable to specifically attribute the exogenous variation in education, whereas OLS is likely biased due to unobserved factors such as ability and family background. However, the IV estimate should be interpreted as a local average treatment effect for individuals whose education is influenced by proximity to a college.

4.5 Adding IQ as a control

The dataset includes `IQ` scores for a subset of respondents.

(a) Run OLS of `lwage` on `educ`, IQ, and all other controls. How does including `IQ` change the OLS coefficient on `educ`?

```r
ols_iq <- lm_robust(lwage ~ educ + IQ + black + smsa + south + exper + expersq + married,
                    data = card, se_type = "HC2")
summary(ols_iq)
```

```
Call:
lm_robust(formula = lwage ~ educ + IQ + black + smsa + south +
    exper + expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept)  4.62482  0.1106112  41.812 7.250e-277  4.407898  4.841742 2051
educ         0.06757  0.0050672  13.334  5.822e-39  0.057631  0.077506 2051
IQ           0.00250  0.0007492   3.337  8.610e-04  0.001031  0.003969 2051
black       -0.12003  0.0270219  -4.442  9.388e-06 -0.173024 -0.067038 2051
smsa         0.16278  0.0184132   8.841  2.006e-18  0.126672  0.198893 2051
south       -0.08318  0.0183564  -4.531  6.195e-06 -0.119180 -0.047182 2051
exper        0.08323  0.0092631   8.985  5.710e-19  0.065065  0.101397 2051
expersq     -0.00235  0.0004643  -5.062  4.519e-07 -0.003261 -0.001440 2051
married     -0.02924  0.0044083  -6.633  4.194e-11 -0.037886 -0.020595 2051

Multiple R-squared:  0.2438 ,   Adjusted R-squared:  0.2409
F-statistic: 85.17 on 8 and 2051 DF,  p-value: < 2.2e-16
```

Including IQ, which we use as a proxy for innate ability that changes the baseline for certain individuals, causes the estimated return to education to decrease from 0.071498 to 0.06757, suggesting that part of the original OLS estimate was in fact picking up variation from innate ability differences.

(b) Now run 2SLS including `IQ`. Does the 2SLS estimate change substantially when `IQ` is added?

```r
iv_iq <- iv_robust(lwage ~ educ + IQ + black + smsa + south + exper + expersq + married |
                     nearc4 + IQ + black + smsa + south + exper + expersq + married,
                   data = card, se_type = "HC2")
summary(iv_iq)
```

```
Call:
iv_robust(formula = lwage ~ educ + IQ + black + smsa + south +
    exper + expersq + married | nearc4 + IQ + black + smsa +
    south + exper + expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
              Estimate Std. Error t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept)  4.1254833   0.803938  5.1316 3.144e-07  2.548863  5.702103 2051
educ         0.1093899   0.066961  1.6336 1.025e-01 -0.021928  0.240708 2051
IQ           0.0004229   0.003414  0.1239 9.014e-01 -0.006271  0.007117 2051
black       -0.1233785   0.028042 -4.3997 1.140e-05 -0.178373 -0.068384 2051
smsa         0.1536399   0.023901  6.4281 1.603e-10  0.106766  0.200513 2051
south       -0.0857686   0.019021 -4.5092 6.873e-06 -0.123070 -0.048467 2051
exper        0.1035065   0.033981  3.0460 2.349e-03  0.036865  0.170148 2051
expersq     -0.0027890   0.000863 -3.2316 1.250e-03 -0.004482 -0.001096 2051
married     -0.0274087   0.005300 -5.1713 2.551e-07 -0.037803 -0.017014 2051

Multiple R-squared:  0.2161 ,   Adjusted R-squared:  0.213
F-statistic: 59.85 on 8 and 2051 DF,  p-value: < 2.2e-16
```

Again, we see that including IQ causes the estimated return to education to decrease, now from 0.124893 to 0.1093899.

(c) What does the comparison between part (a) and part (b) suggest about the source of the discrepancy between OLS and IV? The comparison suggests that ability bias is not the primary reason for the gap between OLS and IV, and instead, the difference is more likely driven by measurement error in education or the fact that IV identifies a local average treatment effect for individuals with relatively high returns to schooling.

5 Interpretation and LATE (15 points)

5.1 Who are the compliers?

In a randomized experiment, the ATE averages over *everyone.* But IV estimates a **Local Average Treatment Effect (LATE)** — the average effect for **compliers**: units whose treatment status changes when the instrument changes. In this context, who are the compliers? Describe them in concrete terms: what type of person's educational attainment was increased because they happened to grow up near a 4-year college?

In this context, compliers are individuals whose years of education increased because they grew up near a 4-year college. These are people who would not have attended college or would have completed less schooling if they had lived farther away, but did obtain more education because proximity reduced the cost of attending. They are likely to be individuals on the margin of attending college, so their decisions are affected by other factors such as access and cost.

5.2 Write the LATE interpretation

Write a single careful sentence interpreting the 2SLS coefficient as a LATE. Your sentence should specify:

(i) The causal effect being measured

(ii) For which specific group of people

(iii) What source of variation identifies it

The 2SLS estimate measures the causal effect of an additional year of education on log weekly wages for individuals whose years of education completed was increased by growing up near a 4-year college (compliers), where the effect is identified using variation in education induced by geographic proximity to a college (`nearc4`).

5.3 External validity

Card finds a 2SLS estimate around 12–13% return to each additional year of education. Is this LATE likely to be the same as the ATE (the return for the average person in the sample)? Think about whether the compliers — men who went to more school because they happened to live near a college — are likely to have *higher* or *lower* returns to education than the typical person. Explain your reasoning.

The LATE estimated by 2SLS is unlikely to be equal to the ATE for the full sample. The IV estimate applies specifically to compliers, meaning men whose years of education completed increased because they grew up near a 4-year college. These individuals are likely on the margin

of attending college, so their decisions to continue seeking higher education are affected by access and costs. Because of this, their returns to education might be different from those of the average person in that these compliers may experience relatively high returns if proximity enabled individuals with strong potential gains to attend college when they otherwise would not have. So, the IV estimate reflects the return to education for this specific subgroup and may not generalize to the broader population.